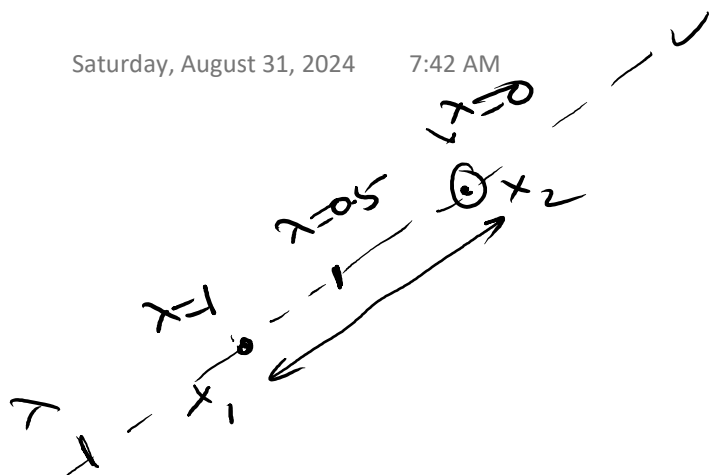


$$\boxed{\lambda x_1 + (1-\lambda)x_2} \quad \checkmark$$

$$\lambda=0 \rightarrow x_2$$

$$\lambda=1 \rightarrow x_1$$

$$\lambda=0.5 \rightarrow \frac{x_1 + x_2}{2}$$

$$\boxed{\lambda \in [0, 1]}$$

λ

$$f(x, y) = 3x^2 + 7xy + 5y^2$$

First order condition:

$$\underline{\underline{\nabla f}} = \begin{pmatrix} f_x \\ f_y \end{pmatrix} = \begin{pmatrix} 6x + 7y \\ 7x + \underline{\underline{10y}} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} x^* \\ y^* \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Second order condition:

$$H = \nabla^2 f = \begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix} = \begin{pmatrix} \textcircled{6} & 7 \\ 7 & 10 \end{pmatrix} \quad D = 60 - 49 = 11 > 0$$

$6 > 0$
(minima)

$$\textcircled{2} \quad f(x, y) = 3x^2 + 8xy + 5y^2$$

$$\nabla f = \begin{pmatrix} f_x \\ f_y \end{pmatrix} = \begin{pmatrix} 6x + 8y \\ 8x + 10y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} x^* \\ y^* \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\nabla^2 f = \begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix} = \begin{pmatrix} 6 & 8 \\ 8 & 10 \end{pmatrix}, \quad D = 60 - 64 = -4 < 0$$

② Max xy
 s.t. $x+y=20$, $x, y \geq 0$

↓ reduce variables

Max $x(20-x)$

$f(x) = x(20-x)$
 $= 20x - x^2$

$f'(x) = 20 - 2x = 0$

$x^* = 10$

$y^* = 10$

$f''(x) = -2 < 0$
 (maxima)

Max xy
 s.t. $x+y=20$
 g: $x+y-20=0$

$\mathcal{L}(x, y, \lambda) = xy + \lambda(x+y-20)$

$\frac{\partial \mathcal{L}}{\partial x} = y + \lambda = 0$

$\frac{\partial \mathcal{L}}{\partial y} = x + \lambda = 0$

$\frac{\partial \mathcal{L}}{\partial \lambda} = x+y-20=0$
 $\Rightarrow -\lambda - \lambda - 20 = 0$
 $\Rightarrow \lambda = -10$

$x^* = 10$, $y^* = 10$

$Z = \{z \in \mathbb{R}^2 : z^T \nabla g(x^*) = 0\}$

$z = \begin{pmatrix} z_1 \\ z_2 \end{pmatrix}$, $\nabla g(x^*) = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$z^T \nabla g(x^*) = \begin{pmatrix} z_1 & z_2 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 0$
 $\Rightarrow z_1 + z_2 = 0$

$\begin{pmatrix} z_1 \\ -z_1 \end{pmatrix}$, $H(\mathcal{L}) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

$z^T H(\mathcal{L}) \cdot z$

$\Rightarrow \begin{pmatrix} z_1 & -z_1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} z_1 \\ -z_1 \end{pmatrix}$

$= \begin{pmatrix} z_1 & -z_1 \end{pmatrix} \begin{pmatrix} -z_1 \\ z_1 \end{pmatrix} = -z_1^2 - z_1^2 = -2z_1^2 < 0$
 maxima

$$\text{Min } z = f(x, y) = 2x^2 + y^2$$

$$\text{s.t. } x + y = 1 \Rightarrow x + y - 1 = 0$$

$$\mathcal{L}(x, y, \lambda) = 2x^2 + y^2 + \lambda(x + y - 1)$$

$$\frac{\partial \mathcal{L}}{\partial x} = 4x + \lambda = 0, \quad x = -\lambda/4$$

$$\frac{\partial \mathcal{L}}{\partial y} = 2y + \lambda = 0, \quad y = -\lambda/2$$

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \lambda} &= x + y - 1 = 0, & -\lambda/4 - \lambda/2 - 1 &= 0 \\ & & \Rightarrow -\lambda - 2\lambda - 4 &= 0 \\ & & \Rightarrow \lambda &= -4/3 \end{aligned}$$

$$x^* = 1/3, \quad y^* = 2/3, \quad z^* = 2\left(\frac{1}{3}\right)^2 + \left(\frac{2}{3}\right)^2 = \frac{2}{3}$$

$$z = \begin{pmatrix} z_1 \\ z_2 \end{pmatrix}, \quad z^T \nabla g(x^*) = 0$$

$$\Rightarrow (z_1, z_2) \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 0 \Rightarrow z_1 + z_2 = 0$$

$$z = \begin{pmatrix} z_1 \\ -z_1 \end{pmatrix}, \quad H(z) = \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}$$

$$z^T H z = (z_1, -z_1) \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} z_1 \\ -z_1 \end{pmatrix}$$

$$= (z_1, -z_1) \begin{pmatrix} 4z_1 \\ -2z_1 \end{pmatrix} = 4z_1^2 + 2z_1^2 > 0 \quad (\text{minima})$$

$$\textcircled{4} \quad \min f(x, y) = x^2 + 2y^2$$

$$\text{s.t.} \quad x^2 + y^2 - 10 = 0.$$

$$\mathcal{L}(x, y, \lambda) = x^2 + 2y^2 + \lambda(x^2 + y^2 - 10) \quad \begin{pmatrix} x^* \\ y^* \\ \lambda^* \end{pmatrix}$$

$$\checkmark \quad \mathcal{L}_x = 2x + 2x\lambda = 0 \Rightarrow \underline{x=0}, \underline{\lambda=-1}$$

$$\checkmark \quad \mathcal{L}_y = 4y + 2y\lambda = 0 \Rightarrow \underline{y=0}, \underline{\lambda=-2}$$

$$\checkmark \quad \mathcal{L}_\lambda = \underline{x^2 + y^2 - 10 = 0.}$$

$$\text{Case 1: } x=0, y=0. \quad \times$$

$$\left\{ \begin{array}{l} \text{Case 2: } x=0, \lambda=-2 \Rightarrow \underline{(0, \sqrt{10})}, \underline{(0, -\sqrt{10})} \\ \text{Case 3: } y=0, \lambda=-1 \Rightarrow \underline{(-\sqrt{10}, 0)}, \underline{(\sqrt{10}, 0)} \end{array} \right.$$

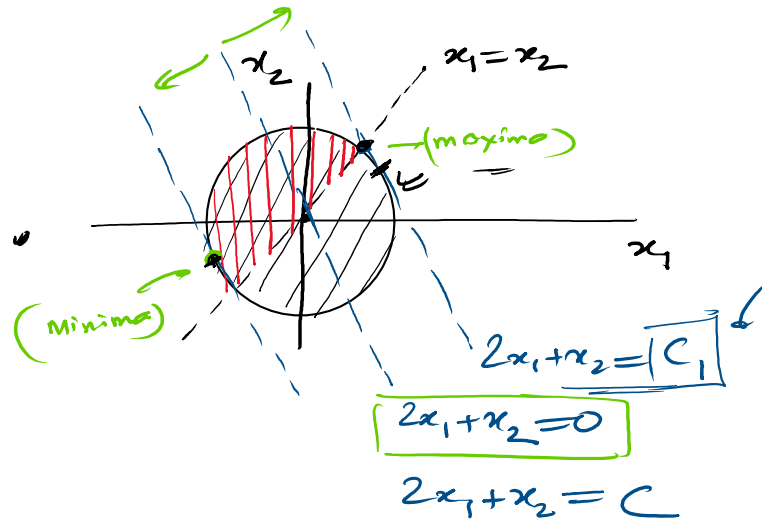
$$\text{Case 4: } \lambda=-1, \lambda=-2 \Rightarrow \times$$

kkT.

Saturday, August 31, 2024

9:26 AM

$$\textcircled{1} \quad \min_{\text{s.t.}} \quad \begin{cases} 2x_1 + x_2 \\ x_1^2 + x_2^2 - 4 \leq 0 \\ x_1 - x_2 \leq 0 \end{cases}$$



$$\textcircled{-2} \quad x^2 + y^2 = 4$$

$$2x + 2y \cdot \frac{dy}{dx} = 0$$

$$\Rightarrow \frac{dy}{dx} = (-x/y) = -2$$

$$(x = 2y) \quad \textcircled{x_1 = 2x_2}$$

$$x_1^2 + x_2^2 = 4$$

$$4x_2^2 + x_2^2 = 4$$

$$x_2 = \pm \frac{2}{\sqrt{5}}, \quad x_1 = \pm \frac{4}{\sqrt{5}}$$

$$\left(-\frac{4}{\sqrt{5}}, -\frac{2}{\sqrt{5}}\right), \quad \left(\frac{4}{\sqrt{5}}, \frac{2}{\sqrt{5}}\right)$$

↓

minimo

$$\textcircled{1} \min f(x) = 2x_1 + x_2$$

$$\text{s.t.} \quad x_1^2 + x_2^2 - 4 \leq 0 \quad (g_1), \lambda_1$$

$$x_1 - x_2 \leq 0 \quad (g_2), \lambda_2$$

$$H(g_1) = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \rightarrow$$

$$\begin{vmatrix} 2-\lambda & 0 \\ 0 & 2-\lambda \end{vmatrix} = (2-\lambda)^2 = 0, \lambda = \underline{2, 2}$$

$$L(x_1, x_2, \lambda_1, \lambda_2) = 2x_1 + x_2 + \lambda_1(x_1^2 + x_2^2 - 4) + \lambda_2(x_1 - x_2)$$

$$\nabla f + \lambda_1 \nabla g_1 + \lambda_2 \nabla g_2 = 0$$

$$\begin{pmatrix} 2 \\ 1 \end{pmatrix} + \lambda_1 \begin{pmatrix} 2x_1 \\ 2x_2 \end{pmatrix} + \lambda_2 \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\left\{ \begin{array}{l} 2 + 2x_1 \lambda_1 + \lambda_2 = 0 \\ 1 + 2x_2 \lambda_1 - \lambda_2 = 0 \end{array} \right\}$$

$$\left\{ \begin{array}{l} \lambda_1(x_1^2 + x_2^2 - 4) = 0 \\ \lambda_2(x_1 - x_2) = 0 \end{array} \right\}$$

$$\psi \left\{ \begin{array}{l} x_1^2 + x_2^2 - 4 \leq 0 \\ x_1 - x_2 \leq 0 \\ \lambda_1, \lambda_2 \geq 0 \end{array} \right\}$$

$$\text{Case 1: } \lambda_1 = 0, \lambda_2 = 0 \quad \times$$

$$\text{Case 2: } \lambda_1 = 0, x_1 = x_2 \quad \times$$

$$\text{Case 3: } x_1^2 + x_2^2 - 4 = 0, \lambda_2 = 0$$

$$\text{Case 4: } x_1^2 + x_2^2 - 4 = 0, x_1 = x_2$$

$$x_1 = -\frac{1}{\lambda_1}, x_2 = -\frac{1}{2\lambda_1}$$

$$\left(-\frac{1}{\lambda_1}\right)^2 + \left(-\frac{1}{2\lambda_1}\right)^2 = 4$$

$$\Rightarrow \frac{1}{\lambda_1^2} + \frac{1}{4\lambda_1^2} = 4$$

$$\Rightarrow \frac{5}{4\lambda_1^2} = 4$$

$$\Rightarrow \lambda_1 = \frac{\sqrt{5}}{4}$$

$$x_1 = -\frac{4}{\sqrt{5}}, x_2 = -\frac{2}{\sqrt{5}}$$